

$$E(1, 3/2, 1, 1/2) = -\frac{7}{8}J_1 + \frac{5}{2}J_2 - J_3 \quad (\text{doublet}) \quad (\text{A.289})$$

$$E(1, 1/2, 1, 3/2) = \frac{5}{8}J_1 - \frac{1}{2}J_2 - J_3 \quad (\text{quadruplet}) \quad (\text{A.290})$$

$$E(1, 1/2, 1, 1/2) = \frac{5}{8}J_1 + J_2 - J_3 \quad (\text{doublet}) \quad (\text{A.291})$$

$$E(0, 1/2, 1, 3/2) = -\frac{3}{8}J_1 - \frac{1}{2}J_2 - J_3 \quad (\text{quadruplet}) \quad (\text{A.292})$$

$$E(0, 1/2, 1, 1/2) = -\frac{3}{8}J_1 + J_2 - J_3 \quad (\text{doublet}) \quad (\text{A.293})$$

$$E(1, 3/2, 0, 3/2) = -\frac{7}{8}J_1 \quad (\text{quadruplet}) \quad (\text{A.294})$$

$$E(1, 1/2, 0, 1/2) = \frac{5}{8}J_1 \quad (\text{doublet}) \quad (\text{A.295})$$

$$E(0, 1/2, 0, 1/2) = -\frac{3}{8}J_1 \quad (\text{doublet}). \quad (\text{A.296})$$

A.8 Chapter 8

Conceptual exercises

Solution 8.1. Let us consider a total Hamiltonian that is the sum of an unperturbed term (diagonal) and a perturbative one (non-diagonal). By diagonalizing this total Hamiltonian using the secular equation and then obtaining its respective vectors, we will observe that these vectors can be written in terms of various vectors of the unperturbed term. In other words, by adding a non-diagonal perturbation, these elements promote a mixing of the unperturbed states. The same happens in perturbation theory, where the first order correction $|n^{(1)}\rangle$ to the state $|n\rangle$ begins to exhibit this mixing of states. In short, the fact that the first order correction $|n^{(1)}\rangle$ to the state $|n\rangle$ depends on other states $|m^{(0)}\rangle$ means that the states mix after the perturbation. In addition, note that, according to Eq. (8.34), the closer these states are in (unperturbed) energy, the stronger the mixing will be, meaning that the dependence of $|n^{(1)}\rangle$ on $|m^{(0)}\rangle$ will be greater.

Text-based exercises

Solution 8.3.

$$\lambda^k : \quad \left(\hat{\mathcal{H}}_0 - E_n^{(0)} \right) |n^{(k)}\rangle = -\hat{\mathcal{V}} |n^{(k-1)}\rangle + \sum_{p=1}^k E_n^{(p)} |n^{(k-p)}\rangle, \quad k \geq 1. \quad (\text{A.297})$$

Note that $k = 0$ represents the unperturbed case and is not included in the recurrence above.

Solution 8.4.

$$\lambda^k : \sum_{p=0}^k \langle n^{(p)} | n^{(k-p)} \rangle = 0, \quad k \geq 1. \quad (\text{A.298})$$

Note that $k = 0$ represents the unperturbed case and is not included in the recurrence above.

$$\textbf{Solution 8.6. 1. } [\hat{L}^2, \hat{z}] = 2i\hbar(\hat{x}\hat{L}_y - \hat{L}_x\hat{y}).$$

$$2. [\hat{L}^2, [\hat{L}^2, \hat{z}]] = 2\hbar^2\{\hat{L}^2, \hat{z}\}.$$

Practice exercises

Solution 8.7. First, we must write $x^2 + e^{\epsilon x} - 5 = 0$. For this problem $\epsilon \ll 1$, and therefore:

$$e^{\epsilon x} \approx 1 + \epsilon x + \frac{1}{2}\epsilon^2 x^2. \quad (\text{A.299})$$

Following the perturbation procedure, the zeros of the function can be expressed in powers of ϵ :

$$z = z_0 + \epsilon z_1 + \epsilon^2 z_2. \quad (\text{A.300})$$

From these assumptions and separating the terms with the same power of ϵ we obtain:

$$\epsilon^0 : z_0^2 - 4 = 0 \Rightarrow z_{0\pm} = \pm 2 \quad (\text{A.301})$$

$$\epsilon^1 : 2z_0 z_1 + z_0 = 0 \Rightarrow z_1 = -\frac{1}{2} \quad (\text{A.302})$$

$$\epsilon^2 : 2z_0 z_2 + z_1^2 + z_1 + \frac{z_0^2}{2} = 0 \Rightarrow z_{2\pm} = \mp \frac{7}{16}. \quad (\text{A.303})$$

Thus, substituting the results above into Eq. (A.300), we obtain the solution we are looking for:

$$z_+ = 2 - \frac{\epsilon}{2} - \frac{7}{16}\epsilon^2 \quad \text{and} \quad z_- = -2 - \frac{\epsilon}{2} + \frac{7}{16}\epsilon^2. \quad (\text{A.304})$$

Solution 8.8. First, let's expand the solution of the problem in powers of the perturbative parameter ϵ . Thus:

$$y(t) = y_0(t) + \epsilon y_1(t) + \epsilon^2 y_2(t). \quad (\text{A.305})$$

The initial condition $y(0) = 0$ in the equation above leads to:

$$y(0) = y_0(0) + \epsilon y_1(0) + \epsilon^2 y_2(0) = 0 + \epsilon 0 + \epsilon^2 0. \quad (\text{A.306})$$

By comparing the terms in powers of ϵ , we obtain:

$$y_0(0) = 0, \quad y_1(0) = 0, \quad \text{and} \quad y_2(0) = 0. \quad (\text{A.307})$$

The first derivative of Eq. (A.305) results in:

$$\dot{y}(t) = \dot{y}_0(t) + \epsilon \dot{y}_1(t) + \epsilon^2 \dot{y}_2(t). \quad (\text{A.308})$$

Analogously to the procedure performed above, that is, considering the initial condition $\dot{y}(0) = 1$ and comparing terms in powers of ϵ , we obtain:

$$\dot{y}_0(0) = 1, \quad \dot{y}_1(0) = 0, \quad \text{and} \quad \dot{y}_2(0) = 0. \quad (\text{A.309})$$

The second derivative of Eq. (A.305) results in:

$$\ddot{y}(t) = \ddot{y}_0(t) + \epsilon \ddot{y}_1(t) + \epsilon^2 \ddot{y}_2(t). \quad (\text{A.310})$$

What we need to do now is substitute Eqs. (A.305), (A.308), and (A.310) into the differential equation of the problem:

$$\ddot{y}(t) + \epsilon \dot{y}(t) + 1 = 0. \quad (\text{A.311})$$

After this procedure, we can compare the terms in powers of ϵ :

$$\epsilon^0: \quad \ddot{y}_0(t) = -1 \quad \Rightarrow \quad y_0(t) = -\frac{t^2}{2} + at + b. \quad (\text{A.312})$$

However, if we use the conditions from Eqs. (A.307) and (A.309), we conclude that $b = 0$ and $a = 1$. Thus:

$$y_0(t) = -\frac{t^2}{2} + t. \quad (\text{A.313})$$

Proceeding further, let's move on to the higher orders in ϵ :

$$\epsilon^1: \quad \ddot{y}_1(t) + \dot{y}_0(t) = 0 \quad \Rightarrow \quad \ddot{y}_1(t) = t - 1. \quad (\text{A.314})$$

Solving the equation above and once again considering the conditions from Eqs. (A.307) and (A.309), we obtain:

$$y_1(t) = \frac{t^3}{6} - \frac{t^2}{2}. \quad (\text{A.315})$$

Similarly, for the second-order correction in ϵ :

$$\epsilon^2: \quad \ddot{y}_2(t) + \dot{y}_1(t) = 0 \quad \Rightarrow \quad \ddot{y}_2(t) = t - \frac{t^2}{2}. \quad (\text{A.316})$$

Solving the equation above and once again considering the conditions from Eqs. (A.307) and (A.309), we obtain:

$$y_2(t) = -\frac{t^4}{24} + \frac{t^3}{6}. \quad (\text{A.317})$$

Finally, the solution to the differential equation of the problem, up to second order in ϵ , is summarized as:

$$y(t) = \left(t - \frac{t^2}{2}\right) + \epsilon \left(\frac{t^3}{6} - \frac{t^2}{2}\right) + \epsilon^2 \left(\frac{t^3}{6} - \frac{t^4}{24}\right). \quad (\text{A.318})$$

Development exercises

Solution 8.9. To proceed with this problem, we need the unperturbed solution. In this case, the wave function is given by:

$$\Psi_n^{(0)}(x) = \sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right), \quad (\text{A.319})$$

and the energy spectrum given by:

$$E_n^{(0)} = \frac{\hbar^2}{2m} \frac{\pi^2}{a^2} n^2. \quad (\text{A.320})$$

Above, $n = 1, 2, 3, \dots$, and we can easily observe that the system is non-degenerate. The first order energy correction is given by:

$$\begin{aligned} E_n^{(1)} &= \langle n^{(0)} | \hat{V} | n^{(0)} \rangle \\ &= \int_0^a \sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right) a\omega_0 \delta(x - a/2) \sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right) dx \\ &= 2\omega_0 \sin^2\left(n\frac{\pi}{2}\right), \end{aligned} \quad (\text{A.321})$$

which can be summarized as:

$$E_n^{(1)} = \begin{cases} 0, & n \text{ par}, \\ 2\omega_0, & n \text{ impar}. \end{cases} \quad (\text{A.322})$$

2. For the second order energy correction, we use Eq. (8.48), and for that, we need the following matrix elements:

$$\begin{aligned} \langle m^{(0)} | \hat{V} | n^{(0)} \rangle &= \int_0^a \sqrt{\frac{2}{a}} \sin\left(m\pi \frac{x}{a}\right) a\omega_0 \delta(x - a/2) \sqrt{\frac{2}{a}} \sin\left(n\pi \frac{x}{a}\right) dx \\ &= 2\omega_0 \sin\left(m\frac{\pi}{2}\right) \sin\left(n\frac{\pi}{2}\right). \end{aligned} \quad (\text{A.323})$$

We should now substitute the above result into Eq. (8.48), and thus:

$$E_1^{(2)} = \sum_{m=2}^{\infty} \frac{4\omega_0^2}{\alpha} \frac{\sin^2\left(m\frac{\pi}{2}\right)}{(1-m^2)}, \quad (\text{A.324})$$

where (i) $\alpha = \hbar^2\pi^2/2ma^2$ and (ii) since the summation in Eq. (8.48) must be for all values of m different from n , we necessarily have $m \geq 2$. The summation is convergent and has an analytical solution [18]. Thus:

$$E_1^{(2)} = -\frac{2ma^2\omega_0^2}{\hbar^2\pi^2}. \quad (\text{A.325})$$

Solution 8.10. 1. The Hamiltonian of the problem can be written as:

$$\hat{\mathcal{H}} = g\mu_B(\hat{S}_z B_z + \hat{S}_x B_x), \quad (\text{A.326})$$

or, in a similar fashion:

$$\begin{aligned} \hat{\mathcal{H}} &= b_z(\hat{S}_z + \lambda\hat{S}_x) \\ &= \frac{b_z}{2} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} + \lambda \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right] \end{aligned} \quad (\text{A.327})$$

where $\lambda = B_x/B_z$ and $b_z = g\mu_B B_z$. Using the matrices above and Eqs. (8.33) and (8.48), we can obtain the energy corrections for the system, as presented in Table A.10. Similarly, with the matrices above and Eqs. (8.34) and (8.49), we can obtain the vector corrections, which are shown in the same table.

TABLE A.10 Energy and vector corrections for the Exercise 8.10.

n	$E_n^{(0)}$	$E_n^{(1)}$	$E_n^{(2)}$	$ n^{(0)}\rangle$	$ n^{(1)}\rangle$	$ n^{(2)}\rangle$
1	$b_z/2$	0	$b_z/4$	$ 1^{(0)}\rangle$	$(1/2) 2^{(0)}\rangle$	$-(1/8) 1^{(0)}\rangle$
2	$-b_z/2$	0	$-b_z/4$	$ 2^{(0)}\rangle$	$-(1/2) 1^{(0)}\rangle$	$-(1/8) 2^{(0)}\rangle$

2. Let's consider the total Hamiltonian of the problem:

$$\hat{\mathcal{H}} = \frac{b_z}{2} \begin{pmatrix} 1 & \lambda \\ \lambda & -1 \end{pmatrix}. \quad (\text{A.328})$$

To obtain the energy eigenvalues and their respective vectors (for comparison with the previous item), we need to solve the secular equation, which leads us to the following eigenvalues:

$$\Gamma_1 = \frac{b_z}{2} \sqrt{\lambda^2 + 1} \xrightarrow{\lambda \rightarrow 0} \frac{b_z}{2} + \frac{b_z}{4} \lambda^2 \quad (\text{A.329})$$

$$\Gamma_2 = -\frac{b_z}{2} \sqrt{\lambda^2 + 1} \xrightarrow{\lambda \rightarrow 0} -\frac{b_z}{2} - \frac{b_z}{4} \lambda^2. \quad (\text{A.330})$$

Notice, therefore, that the eigenvalues presented above, in the case $\lambda \rightarrow 0$, recover the results from item (1). Continuing, we need to obtain the vectors, a task that must also be performed using the secular equation. Thus:

$$|1\rangle = \begin{pmatrix} \alpha_+ \\ \beta_+ \end{pmatrix} \quad \text{and} \quad |2\rangle = \begin{pmatrix} \alpha_- \\ \beta_- \end{pmatrix} \quad (\text{A.331})$$

where

$$\alpha_{\pm} = \pm \frac{1}{\sqrt{2}} \frac{\sqrt{\lambda^2 + 1} \pm 1}{[\lambda^2 + 1 \pm \sqrt{\lambda^2 + 1}]^{1/2}} \quad (\text{A.332})$$

and

$$\beta_{\pm} = \frac{1}{\sqrt{2}} \frac{\lambda}{[\lambda^2 + 1 \pm \sqrt{\lambda^2 + 1}]^{1/2}}. \quad (\text{A.333})$$

By performing the expansion for $\lambda \rightarrow 0$, we obtain:

$$|1\rangle = |1^{(0)}\rangle + \frac{\lambda}{2}|2^{(0)}\rangle - \frac{\lambda^2}{8}|1^{(0)}\rangle \quad (\text{A.334})$$

and

$$|2\rangle = |2^{(0)}\rangle - \frac{\lambda}{2}|1^{(0)}\rangle - \frac{\lambda^2}{8}|2^{(0)}\rangle, \quad (\text{A.335})$$

which indeed recovers the results obtained in item (1).

Advanced exercises

Solution 8.11. The Hamiltonian of the problem is given by:

$$\hat{\mathcal{H}} = D\hat{S}_{1z}\hat{S}_{2z} + g\mu_B B_x \hat{S}_x, \quad (\text{A.336})$$

where the first term represents the unperturbed contribution and the second term the perturbation, with $\lambda = B_x$ (according to van Vleck's formalism). After expanding the Hilbert space, we can write:

$$\hat{\mathcal{H}}_0 = \frac{D}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \hat{\mathcal{W}} = \frac{g\mu_B}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}. \quad (\text{A.337})$$

These matrices are written in the local basis, given by:

$$|++\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |+-\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |-+\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |--\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (\text{A.338})$$

Observing the degeneracy in $\hat{\mathcal{H}}_0$, we can write the projection operators $\hat{\mathbb{A}}$ and $\hat{\mathbb{B}}$. These are:

$$\hat{\mathbb{A}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \hat{\mathbb{B}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\text{A.339})$$

whereas the B -reduced and A -reduced perturbation matrices read as $\hat{\mathcal{W}}_B = \hat{\mathcal{W}}_A = 0$. From the results up to this point, it is possible to write (using Eqs. (8.96) and (8.97)) the zero order energy:

$$E_{1a}^{(0)} = E_{2a}^{(0)} = \frac{D}{4} \quad \text{and} \quad E_{1b}^{(0)} = E_{2b}^{(0)} = -\frac{D}{4}, \quad (\text{A.340})$$

and the first order energy correction:

$$E_{1a}^{(1)} = E_{2a}^{(1)} = E_{1b}^{(1)} = E_{2b}^{(1)} = 0. \quad (\text{A.341})$$

To write the second order energy correction, we need to recall Eq. (8.130). After simple calculations, we obtain:

$$E_{1a}^{(2)} = E_{2a}^{(2)} = \frac{(g\mu_B)^2}{D} \quad \text{and} \quad E_{1b}^{(2)} = E_{2b}^{(2)} = -\frac{(g\mu_B)^2}{D}. \quad (\text{A.342})$$

By using Eq. (8.A.11), the van Vleck susceptibility results in:

$$\frac{\chi_{xx}}{N\mu_0} = 2 \frac{(g\mu_B)^2}{D} \tanh\left(\frac{D}{4k_B T}\right). \quad (\text{A.343})$$

Solution 8.12. Considering the matrix representation of each operator in the local basis (Eq. (A.338)), we can write:

$$\hat{\mathcal{H}}_0 = -\frac{J}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \hat{\mathcal{W}} = \frac{g\mu_B}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \quad (\text{A.344})$$

where $\lambda = B_x$. Perturbation theory requires that the unperturbed term be diagonal, and for this reason, we must diagonalize this matrix. We know

the basis change matrix for this system (see Example 7.6) and, by applying the same procedure, we obtain the diagonal unperturbed term. Note, however, that we must also change the perturbation $\hat{\mathcal{W}}$ to the same basis, as the formalism requires both matrices to be in the same basis. Therefore:

$$\hat{\mathcal{H}}'_0 = -\frac{J}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \quad \text{and} \quad \hat{\mathcal{W}}' = \frac{\sqrt{2}g\mu_B}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{A.345})$$

being the new vectors:

$$|1, +1\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |1, 0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |1, -1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |0, 0\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (\text{A.346})$$

Observing the degeneracy in $\hat{\mathcal{H}}'_0$, it is possible to write the B -reduced and A -reduced perturbation matrices. These are:

$$\hat{\mathcal{W}}_A = \frac{\sqrt{2}g\mu_B}{2} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{A.347})$$

and $\hat{\mathcal{W}}_B = 0$. The unperturbed energies are:

$$E_{1a}^{(0)} = E_{2a}^{(0)} = E_{3a}^{(0)} = -\frac{J}{4} \quad \text{and} \quad E_{1b}^{(0)} = \frac{3J}{4}, \quad (\text{A.348})$$

while the first energy correction can be written as:

$$E_{1a}^{(1)} = g\mu_B, \quad E_{2a}^{(1)} = 0, \quad E_{3a}^{(1)} = -g\mu_B, \quad \text{and} \quad E_{1b}^{(1)} = 0. \quad (\text{A.349})$$

Let us now proceed to the second order correction, for which we use Eq. (8.130). After simple calculations, we find that the second order energy correction for all states is zero. Finally, using Eq. (8.A.11), we can express the van Vleck susceptibility for this problem:

$$\frac{\chi_{xx}}{N\mu_0} = 2 \frac{(g\mu_B)^2}{k_B T} \frac{1}{3 + e^{-J/k_B T}}. \quad (\text{A.350})$$

Solution 8.13. 1. After solving the Schrödinger equation, the normalized and unperturbed wave function is given by:

$$\Psi(x, y, z) = \sqrt{\frac{8}{a^3}} \sin\left(\frac{n_x \pi}{a} x\right) \sin\left(\frac{n_y \pi}{a} y\right) \sin\left(\frac{n_z \pi}{a} z\right), \quad (\text{A.351})$$

while the energy spectrum reads as:

$$E_{n_x, n_y, n_z}^{(0)} = \frac{\hbar^2 \pi^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2), \quad (\text{A.352})$$

where $(n_x, n_y, n_z) = 1, 2, 3, \dots$. In addition: $k_u = n_u \pi / a$, where $u = (x, y, z)$. The statement suggests that the problem is limited to the sphere $k_x^2 + k_y^2 + k_z^2 \leq 11\pi^2/a^2$, that is, $n_x^2 + n_y^2 + n_z^2 \leq 11$. Table A.11 presents the possible combinations of values for the quantum numbers and their respective energy values, limited to the mentioned sphere.

TABLE A.11 Possible combinations for the quantum numbers of the Exercise 8.13, as well as their corresponding energies (in units of $\hbar^2 \pi^2 / 2ma^2$).

n_x	n_y	n_z	$E_{n_x, n_y, n_z}^{(0)}$
1	1	1	3
1	1	2	6
1	2	1	6
2	1	1	6
2	2	1	9
2	1	2	9
1	2	2	9
1	1	3	11
1	3	1	11
3	1	1	11

2. The selection rules are the conditions imposed to the quantum numbers involved in such a way that the elements of the perturbation matrix are non-zero. In other words, the following condition must be imposed:

$$\langle n'_x, n'_y, n'_z | \hat{\mathcal{W}} | n_x, n_y, n_z \rangle \neq 0. \quad (\text{A.353})$$

The selection rule is the equation or condition imposed on n'_u and n_u in such a way that the above equation is satisfied. Thus, it is easy to demonstrate that:

$$\begin{aligned} & \langle n'_x, n'_y, n'_z | \hat{\mathcal{W}} | n_x, n_y, n_z \rangle \\ &= \omega_0 \sin\left(\frac{n'_x}{2}\pi\right) \sin\left(\frac{n_x}{2}\pi\right) \sin\left(\frac{n'_y}{2}\pi\right) \sin\left(\frac{n_y}{2}\pi\right) \sin\left(\frac{n'_z}{2}\pi\right) \sin\left(\frac{n_z}{2}\pi\right). \end{aligned} \quad (\text{A.354})$$

We can then see that both (n'_x, n'_y, n'_z) and (n_x, n_y, n_z) need to be odd in order to satisfy Eq. (A.353).

3. From the selection rules, it is possible to construct the perturbation matrix, where the non-zero elements need to be solved case-by-case. However, it is a relatively simple task. The perturbation matrix reads then as:

$$\hat{\mathcal{W}} = \begin{pmatrix} \omega_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\omega_0 & -\omega_0 & -\omega_0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\omega_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \omega_0 & \omega_0 & \omega_0 \\ -\omega_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \omega_0 & \omega_0 & \omega_0 \\ -\omega_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \omega_0 & \omega_0 & \omega_0 \end{pmatrix} \quad (\text{A.355})$$

written in the following basis:

$$\begin{aligned} |111\rangle &= \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & |112\rangle &= \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & |121\rangle &= \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & |211\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & |221\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\ |212\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & |122\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} & |113\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} & |131\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} & |311\rangle &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \end{aligned} \quad (\text{A.356})$$

5. This is a degenerate system, and for this reason, let's write the unperturbed matrix (in the same basis):

$$\hat{\mathcal{H}}_0 = \frac{\hbar^2 \pi^2}{2ma^2} \begin{pmatrix} 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 11 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 11 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 11 & 0 \end{pmatrix} \quad (\text{A.357})$$

The reduced matrices for this problem are given by:

$$\hat{\mathcal{W}}_A = \begin{pmatrix} \omega_0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\hat{\mathcal{W}}_D = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \omega_0 & \omega_0 & \omega_0 & \omega_0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \omega_0 & \omega_0 & \omega_0 & \omega_0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \omega_0 & \omega_0 & \omega_0 & \omega_0 \end{pmatrix} \quad (\text{A.358})$$

and $\hat{\mathcal{W}}_B = \hat{\mathcal{W}}_C = \mathbb{0}$. The degeneracy domains A , B , C , and D follow the energy-ascending order on Table A.11. Thus, it is possible to write the first order correction for the energy as $E_{1a}^{(1)} = \omega_0$, $E_{nb}^{(1)} = E_{nc}^{(1)} = 0$, and, finally, $E_{1d}^{(1)} = E_{2d}^{(1)} = 0$ and $E_{3d}^{(1)} = 3\omega_0$. We conclude, therefore, that this perturbation only lifts the degeneracy of the nd levels.

Solution 8.14. 1. In this case, we aim to determine which elements of the perturbative matrix are non-zero. The problem states that $[\hat{L}_z^2, \hat{\mathcal{O}}] = 0$, and

this commutation leads to:

$$\langle n', l', m' | (\hat{L}_z^2 \hat{O} - \hat{O} \hat{L}_z^2) | n, l, m \rangle = 0 \Rightarrow (m'^2 - m^2) \hbar^2 \langle n', l', m' | \hat{O} | n, l, m \rangle = 0. \quad (\text{A.359})$$

Therefore, for the perturbative matrix elements $\langle n', l', m' | \hat{O} | n, l, m \rangle$ be non-zero, the condition below must be true:

$$m' = \pm m, \quad (\text{A.360})$$

this being the selection rule for the quantum number m . The second commutation provided in the problem, $[\hat{L}^2, [\hat{L}^2, \hat{O}]] = 0$ allows us to obtain the selection rule for the quantum number l :

$$\langle n', l', m' | (\hat{L}^4 \hat{O} - 2\hat{L}^2 \hat{O} \hat{L}^2 + \hat{O} \hat{L}^4) | n, l, m \rangle = 0, \quad (\text{A.361})$$

which, after some algebra, leads to:

$$[l'(l' + 1) - l(l + 1)]^2 \langle n', l', m' | \hat{O} | n, l, m \rangle = 0. \quad (\text{A.362})$$

From the above equation, we obtain the selection rule for the quantum number l :

$$l' = l. \quad (\text{A.363})$$

2. The problem requires us to extend the analysis up to the $n = 2$ level. To achieve this, we will consider the vectors introduced during the study of the Stark Effect (Eq. (8.B.5)), as well as the unperturbed Hamiltonian (Eq. (8.B.6)). Thus, the perturbation matrix reads as:

$$\hat{\mathcal{W}} = \begin{pmatrix} a & u & 0 & 0 & 0 \\ u^* & b & 0 & 0 & 0 \\ 0 & 0 & c & 0 & v \\ 0 & 0 & 0 & d & 0 \\ 0 & 0 & v^* & 0 & f \end{pmatrix}. \quad (\text{A.364})$$

Solution 8.15. First, let's write the unperturbed Hamiltonian $\hat{\mathcal{H}}_0$ and the corresponding basis vector. Considering it must be diagonal, the problem must be written in the coupled basis $|s_1, s_2, s, m\rangle = |s, m\rangle$, given by Eq. (A.346). The unperturbed Hamiltonian was written in Eq. (A.345) and reads as:

$$\hat{\mathcal{H}}_0 = -\frac{J}{4} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}. \quad (\text{A.365})$$

1. Selection rule for s : Note that $[\hat{S}^2, \hat{\mathcal{W}}] = 0$, since $[\hat{S}^2, \hat{S}_\pm] = 0$. Thus:

$$\begin{aligned}\langle s', m' | [\hat{S}^2, \hat{\mathcal{W}}] | s, m \rangle &= 0 \\ \langle s', m' | (\hat{S}^2 \hat{\mathcal{W}} - \hat{\mathcal{W}} \hat{S}^2) | s, m \rangle &= 0 \\ [s'(s' + 1) - s(s + 1)] \hbar^2 \langle s', m' | \hat{\mathcal{W}} | s, m \rangle &= 0.\end{aligned}\quad (\text{A.366})$$

For the perturbative matrix element $\langle s', m' | \hat{\mathcal{W}} | s, m \rangle$ be non-zero, the term $[s'(s' + 1) - s(s + 1)]$ must necessarily be zero. This leads to the selection rule $s' = s$.

Selection rule for m : For this case we need a null operator that includes both $\hat{S}_z$ and $\hat{W}$. It's easy to verify that $[\hat{S}_z, \hat{\mathcal{W}}] = \lambda \hbar (\hat{S}_+^2 - \hat{S}_-^2)$. However, this result is not useful, as we need a null operator containing only $\hat{S}_z$ and $\hat{\mathcal{W}}$. An alternative is to compute $[\hat{S}_z, [\hat{S}_z, \hat{\mathcal{W}}]]$, which can be easily verified to yield the following solution: $[\hat{S}_z, [\hat{S}_z, \hat{\mathcal{W}}]] - 4\hbar^2 \hat{\mathcal{W}} = 0$. Thus:

$$\begin{aligned}\langle s', m' | (\hat{S}_z^2 \hat{\mathcal{W}} - 2\hat{S}_z \hat{\mathcal{W}} \hat{S}_z + \hat{\mathcal{W}} \hat{S}_z^2 - 4\hbar^2 \hat{\mathcal{W}}) | s, m \rangle &= 0 \\ [m'^2 - 2m'm + m^2 - 4] \hbar^2 \langle s', m' | \mathcal{W} | s, m \rangle &= 0\end{aligned}\quad (\text{A.367})$$

which leads us to $[m'^2 - 2m'm + m^2 - 4] = 0$. We can solve this quadratic equation with respect to m' , resulting in the selection rule $m' = m \pm 2$.

2. The perturbative matrix is written considering the basis vector $|s, m\rangle$, described in Eq. (A.346). It results in:

$$\hat{\mathcal{W}} = \begin{pmatrix} 0 & 0 & a^* & 0 \\ 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{A.368})$$

3. The only element we need to determine is the parameter:

$$a = \langle 1, -1 | \hat{\mathcal{W}} | 1, +1 \rangle = \frac{\lambda}{2} \langle 1, -1 | (\hat{S}_+^2 + \hat{S}_-^2) | 1, +1 \rangle = \lambda. \quad (\text{A.369})$$

4. Observing the degeneracy in $\hat{\mathcal{H}}_0$, it is possible to write the B -reduced and A -reduced perturbation matrices. These are:

$$\hat{\mathcal{W}}_A = \lambda \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{A.370})$$

and $\hat{\mathcal{W}}_B = 0$. The unperturbed energies are:

$$E_{1a}^{(0)} = E_{2a}^{(0)} = E_{3a}^{(0)} = -\frac{J}{4} \quad \text{and} \quad E_{1b}^{(0)} = \frac{3J}{4}, \quad (\text{A.371})$$

while the first energy correction can be written as:

$$E_{1a}^{(1)} = \lambda, \quad E_{2a}^{(1)} = 0, \quad ; E_{3a}^{(1)} = -\lambda \quad \text{and} \quad E_{1b}^{(1)} = 0. \quad (\text{A.372})$$

Let us now proceed to the second order correction, for which we use Eq. (8.130). After simple calculations, we find that the second order energy correction for all states is zero. Finally, we can write Table A.12 with the results obtained above.

TABLE A.12 Energy corrections for Exercise 8.15.

Vector	$E_{na/nb}^{(0)}$	$E_{na/nb}^{(1)}$	$E_{na/nb}^{(2)}$
$ 1a\rangle$	$-J/4$	$+\lambda$	0
$ 2a\rangle$	$-J/4$	0	0
$ 3a\rangle$	$-J/4$	$-\lambda$	0
$ 1b\rangle$	$3J/4$	0	0

A.9 Chapter 9

Conceptual exercises

Solution 9.1. The summation (over m) represents all possible intermediate transitions $|m\rangle$ between the initial state $|i\rangle$ and the final state $|f\rangle$.

Text-based exercises

Solution 9.2.

$$[\hat{\mathcal{H}}_0, e^{-i\hat{\mathcal{H}}_0 t/\hbar}] = \sum_{n=0}^{\infty} \left(-\frac{it}{\hbar}\right)^n \frac{1}{n!} [\hat{\mathcal{H}}_0, \hat{\mathcal{H}}_0^n] = 0 \quad (\text{A.373})$$

since $[\hat{\mathcal{H}}_0, \hat{\mathcal{H}}_0^n] = 0$.

Solution 9.3.

$$\begin{aligned}
 e^{\pm i\hat{\mathcal{H}}_0 t/\hbar} |k\rangle &= \sum_{n=0}^{\infty} \left(\pm \frac{it}{\hbar}\right)^n \frac{1}{n!} \hat{\mathcal{H}}_0^n |k\rangle \\
 &= \sum_{n=0}^{\infty} \left(\pm \frac{it}{\hbar}\right)^n \frac{1}{n!} (E_k^{(0)})^n |k\rangle \\
 &= e^{\pm iE_k^{(0)} t/\hbar} |k\rangle.
 \end{aligned} \quad (\text{A.374})$$